

Implementing at the Beamline

The Goal

- One notebook that runs the full XTEC pipeline on any CHESS dataset
- Change one thing (the data path) -> runs top to bottom
- Built and tested it on three materials
 - Ta₂NiSeS₇
 - HfV₂
 - KMnGeOH (with Suchi)

Sections

1. Stacking
 - a. Need to change the temps
2. Load
 - a. Load the stacked file
3. Threshold
 - a. Mask_Zeros, Threshold_Background, Rescale_mean
 - b. Log intensity distribution and Q-space map
4. Sweep + Judge
5. How to run the complete steps
6. Put the whole code in /nfs/chess/id4baux/summer_project/qm2_XTEC

Step 1: Stacking

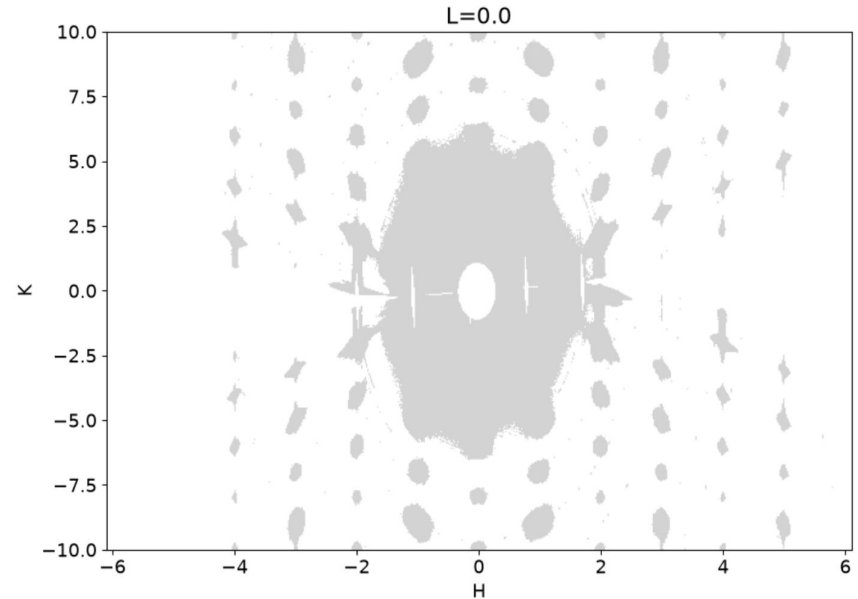
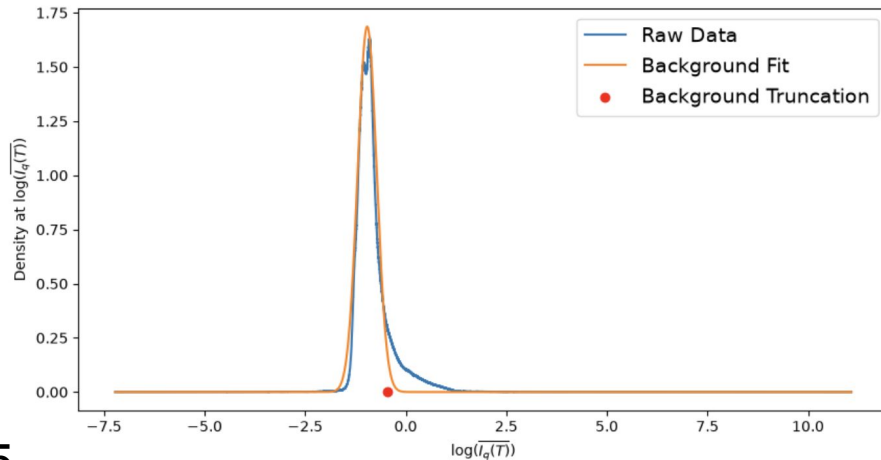
- Raw data comes as separate files, one per temperatures
- The code discovers temperature folders
 - Skips any temperature missing its data file
- Down samples each volume (STEP factor) and combines them into one 4D file: (temp, Ql, Qk, Qh)
 - If needed
- Output: a single {MATERIAL}_stacked.nxs in XTEC format

Step 2: Loading

- Looks at the reciprocal-space ranges first, so you know what region exists
- Loads a chosen slice
- The slice is set once and shared with the sweep, so they always match
- Reads intensity I , the Q-axes, and temperatures into memory for processing

Step 3: Threshold

- Thresholding from XTEC removes low-intensity background
- Cutoff is estimated automatically
- Outputs diagnostic plots
 - Log intensity distribution
 - Q-space map



Slide 4: Sweep + Judge

- Runs XTEC-d clustering across many cluster counts ($k=2-8$) and rescaling methods
 - Mean and z-score
- Judge scores each result and suggest a best k
 - Note: the judges pick isn't always right
- Saves each runs trajectory plots and recommendation.json
- Output is organized into one {MATERIAL}/ folder, one sub folder per rescale

Step 5: How to Run the Code

- The top cell is the only cell where you change things
 - BASE
 - MATERIAL
 - Rescale methods
 - Amount of clusterings
- Run the rest as is

Step 6: Where the Code Lives

- Add beamline implementation into `/nfs/chess/id4baux/summer_project/qm2_XTEC` so others can use it